

Five String Acoustic Resonator Rev B

Revised passive acoustic design and measurement plan • 5 September 2026

Rev B corrects the anchor layout, vessel fit, load review, and acoustic measurements while retaining the Rev A design concept. Status: PROVISIONAL. The nominal envelope is a planning reference; actual measurements and connection calculations are required before cutting or tensioning.

Build intent	Hard boundary	First-pass criterion
Upright five-string resonator using a vessel as an acoustic enclosure and a central hardwood load path.	No high voltage, energized helix, “Tesla coil,” exposed mains, smoke, salt, or mechanism inferred from mythology.	Dimensionally compatible, serviceable, and measurable. Build release follows the completed worksheet and load review.

1 Geometry and load path

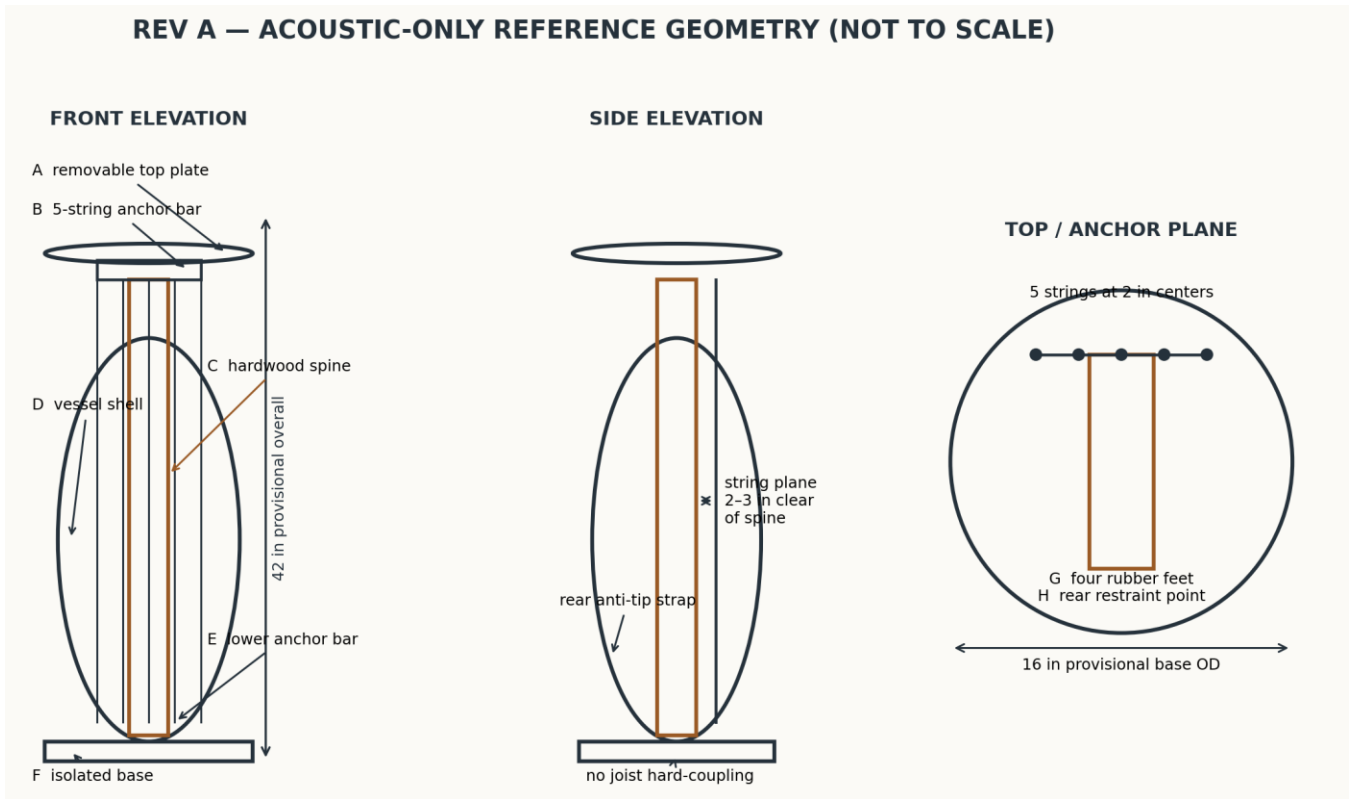


Figure 1. Retained Rev A concept sketch, not a Rev B cutting drawing. Its five-string layout remains illustrative. The corrected fit and anchor rules below govern dimension selection.

Strings occupy one serviceable plane in front of the spine. The upper and lower anchors transfer string forces into the hardwood frame and its connections. The vessel is independently supported and carries no string tension. Freeze the actual spine orientation, string-plane offset, anchor positions, and vessel cross-sections before fabrication.

2 Provisional dimensions and acceptance limits

Feature	Rev B nominal	Allowed / gate	Design note
Overall height	42 in (1067 mm)	36–48 in	Must fit room and remain tip-stable
Base outside diameter	16 in (406 mm)	≥ 14 in	Wider if total center of gravity is high
Vessel maximum OD / clear ID	Measured profile	Check at every height	Do not assume 12 in clear ID is sufficient; include mouth, hardware, and motion.
Top plate diameter	16 in	14–18 in	Thin metal or 1/4–3/8 in plywood; removable
Spine	Nominal 2 × 3 × 40 in	Record actual stock	Declare orientation; review compression, bending, stability, and connections.
Speaking string length L	32 in (813 mm)	28–36 in	All five equal within 1/16 in unless intentionally scaled
String spacing	2 in centers	1.5–2.25 in	Enough access for plucking and damping
String-to-spine clearance	2–3 in	≥ 1.5 in	Measure from the spine face and convert to axis offset for the vessel-fit calculation.
Top acoustic gap	1.5 in	1–2 in	Open annulus; no pinching of airflow
Foot pattern	4 feet on 13–14 in circle	As base permits	Use the convex hull of actual foot contacts for stability, not the base diameter.
Initial tension limit	≤ 20 lbf/string; ≤ 100 lbf total	Planning ceiling only	Use the lower reviewed operating limit; no verified component capacity is implied.

Rev B correction. The old eight-inch anchor bar could not contain the five-hole pattern with positive end allowances. Final bar length and vessel fit must be selected together.

3 Parts list

ID	Part	Qty	Specification	Safety / assembly note
P1	Base frame	1	3/4 in Baltic-birch plywood, laminated to 1.5 in total, 16 in OD	No cracked, warped, or particle-board base
P2	Central spine	1	Hard maple/ash, nominal 2 × 3 × 40 in	Primary load path
P3	Upper/lower anchor bars	2	Material, thickness, width, and length set by the connection calculation	Five holes at 2 in centers need 8 in between end-hole centers PLUS two end allowances.
P4	Tuning machines	5	Enclosed bass or zither pins rated for chosen strings	Mount to separate head block if needed

ID	Part	Qty	Specification	Safety / assembly note
P5	Strings	5	Low-tension acoustic strings selected by calculated tension	Manufacturer data at exact pitch/scale; maximum 20 lbf only if all lower limits permit.
P6	Bridge / nut strips	2	Hardwood, bone, or acetal; replaceable	Rounded bearing edges
P7	Vessel shell	1	Measured vessel profile; independently supported	Fit the complete hardware/string envelope; keep glass free of concentrated loads.
P8	Top plate	1	Thin cymbal-like metal plate or 1/4–3/8 in plywood, 14–18 in	Insulated from shell; removable
P9	Base feet	4	Neoprene/Sorbothane feet, 1–1.5 in OD, through-bolted	Non-slip; equal height
P10	Vessel clamps	3–4	Padded stainless straps or wooden cleats	No point loading on glass
P11	Anti-tip restraint	1	Rated restraint system selected from the anti-tip load case	Record attachment geometry, slack, and rating basis; do not infer capacity from a generic 200 lbf label.
P12	Guards/covers	Var.	Polycarbonate shields over anchors/tuners	Contain a broken string end
P13	Fasteners	Set	Grade-marked machine bolts, washers, nyloc nuts	No wood screws in primary tension path
P14	Optional pickup	1	Passive piezo/contact microphone	Electrically isolated; removable
P15	Optional exciter — Phase 2 only	1	12–24 VDC, ≤30 W tactile exciter + fused isolated supply	Not installed until acoustic baseline passes

4 Measurements and calculations before fabrication

- Room clear height and the exact footprint available, including walking clearance.
- Vessel: total height, maximum outside diameter, mouth inside/outside diameter, wall thickness, base shape, material, mass, and any chips or cracks.
- Chosen top plate/cymbal: diameter, mass, center-hole diameter, edge condition, and mounting hardware.
- Chosen string set: material, gauge/core type, unit weight or manufacturer tension at proposed pitch and 32 in scale, plus safe tuning range.
- Tuner footprint, post diameter, mounting thickness, and required access for winding.
- Spine stock actual dimensions, species, grain direction, straightness, and defects.
- Finished mass, center-of-gravity position, actual foot contact locations, and restraint geometry. Assess static stability before planning a controlled restraint test.
- Wall-stud location for anti-tip restraint; electrical outlet and cable route if Phase 2 is attempted.
- Target pitches and permitted sound-pressure level at the operator position.

Tension gate. Use the exact speaking length, target pitch, and manufacturer string data for each string. The retained planning ceiling is 20 lbf per string and 100 lbf total, or a lower verified component/connection limit. These ceilings are not verified capacities. Record whether each rating is a working-load limit or breaking strength; document the design factor once, against the relevant failure mode. Unknown ratings or unresolved load paths leave the assembly untensioned.

Calculation notes

Anchor layout. Let N be string count, s the hole-center spacing, and e the required end-hole-center allowance. Select e from the actual hole, material, fastener, and load calculation.

$$L_{bar} \geq (N - 1)s + 2e$$

For $N = 5$ and $s = 2$ in, the center span is 8 in. A 10 in bar would leave 1 in center-to-end at each end as geometry only; it is not an approved part size.

Vessel fit. In a centered circular cross-section, let d be actual spine depth toward the string plane, c the gap from its front face to that plane, and δ the wall allowance including hardware and motion. The outer string has sideways offset x . Apply the inequality at every occupied height z .

$$x = \frac{(N - 1)s}{2}, \quad R_{clear}(z) \geq \sqrt{x^2 + (d/2 + c)^2} + \delta$$

With $d = 3$ in, $c = 3$ in, $s = 2$ in, $N = 5$, and $\delta = 1$ in before added hardware or excursion, the required clear diameter is about 14.04 in. This illustrates a failure of the old blanket 12 in rule. Offset or noncircular vessels need actual coordinates instead.

String tension. Use SI units: speaking length L in metres, fundamental f in hertz, mass per length μ in kg/m, and tension T in newtons. For an ideal flexible string at small amplitude:

$$T = 4\mu L^2 f^2$$

At 32 in (0.8128 m), 110 Hz, and 20 lbf (88.9644 N), the corresponding ideal linear density is 2.7823 g/m. This is an example, not a string selection. Doubling pitch with the same string and length quadruples tension; use manufacturer data for the actual string and safe range. Source: UNSW, Strings, standing waves and harmonics (see section 12).

Frame loads. For approximately parallel strings, resolve the resultant axial force and eccentric moment at the section being checked. The offset e_i is measured from that section's reference axis, not automatically from the spine face.

$$P = \sum T_i, \quad M = \sum T_i e_i$$

Review bar bending, spine compression and bending, stability, bolt bearing, joints, and base reactions using actual dimensions and materials. Evaluate restraint demand independently from assembly mass, center of gravity, attachment angle/height, and slack. A 30 min observation cannot establish long-term creep or fatigue capacity.

Static tipping screen. Four equally spaced point feet on a circle of diameter D form a square support polygon. For a rigid assembly with centered mass, no sliding, and no strap contribution:

$$r_{support} = \frac{D}{2\sqrt{2}}, \quad h \tan \theta < r_{support}$$

At $\theta = 10^\circ$ and $D = 13\text{--}14$ in, the centered height limit is about 26.07–28.07 in. Actual foot patches, compliance, offset mass, and restraint engagement change the result. This is a calculation example, not an instruction to tilt an unverified assembly.

Measurement worksheet

Complete the values and review references below before releasing a cutting layout. An empty field is unresolved, not zero.

Record	Measured or selected value	Evidence or review reference
Vessel ID profile and mouth at each height	To record	To record
Actual spine section and orientation	To record	To record
String plane and outer-string coordinates	To record	To record
Anchor hole size, spacing, end allowance	To record	To record
Bar material and calculated connection limits	To record	To record
Frame joints and operating-load limits	To record	To record
Mass and center of gravity in three axes	To record	To record
Foot contacts and stability margin	To record	To record
Restraint geometry and rated load basis	To record	To record
Sensor mounting, excitation, and SPL ceiling	To record	To record

String schedule

String	Product / μ	L (m)	f (Hz)	T (N / lbf)	Limit source
1	To record	To record	To record	To calculate	To record
2	To record	To record	To record	To calculate	To record
3	To record	To record	To record	To calculate	To record
4	To record	To record	To record	To calculate	To record
5	To record	To record	To record	To calculate	To record

Record the total operating tension and the reviewed individual/component limits. Final configuration ID:

Review reference and date: _____

5 Mounting and assembly

1. After the dimensional and structural review is complete, dry-fit the base, spine, upper head block, and lower anchor without the vessel. Use the reviewed through-bolt, backing-plate, and cross-bracing arrangement; record fastener sizes and tightening requirements.
2. Fit the four feet at the documented positions. Confirm contact and no rocking. Install the independently assessed anti-tip restraint with the specified attachment points and slack; the restraint does not validate the string frame.
3. Mount the vessel independently with padded restraints. Verify the clearance envelope at every occupied height, including the mouth, hardware, and planned string excursion. Avoid point loading of glass; never drill tempered glass.
4. Install rounded nut and bridge strips, the dimensioned anchor bars, and guards. Confirm hole edge allowances and tool access against the approved connection layout. Cover cut string ends.
5. Set a target operating tension for each string from the reviewed string schedule. Install and increase tension gradually using the reviewed sequence. Stand out of the string plane and wear eye protection. Watch for bowing, movement, loosening, and loss of clearance.
6. Inspect at approximately 25%, 50%, 75%, and 100% of the reviewed operating tensions, never automatically at the planning ceiling. Stop for cracking, progressive movement, permanent deformation, unstable footing, or vessel contact.
7. Fit the removable top plate on its documented captive mount after the frame is stable. Record how vibration reaches the plate and vessel; incidental proximity alone does not define an acoustic coupling.

6 Airflow and electrical constraints

Domain	Allowed in Rev B	Constraint / exclusion
Airflow	Natural convection; one guarded external fan for mapping only	Fan stays outside vessel, ≥ 12 in from strings and loose fibers. No airflow through electronics; no fan-induced string contact.
Flow visualization	Short yarn tufts or lightweight tissue streamers	No smoke, incense, combustion, powders, salt mist, aerosols, or material that can enter lungs/electronics.

Domain	Allowed in Rev B	Constraint / exclusion
Copper helix	Decorative, mechanically secured, electrically floating	Not required for acoustic operation. It must not touch strings, tuners, pickup wiring, top plate, or electrical ground.
Acoustic sensing	Passive contact mic/piezo into battery recorder	Secure cable strain relief; avoid trip loops.
Powered excitation	Excluded from baseline; Phase 2 may use 12–24 VDC, ≤ 30 W	Isolated listed supply outside vessel, fused at source, guarded connector, no exposed conductors, no mains inside assembly, and never > 48 VDC.
Room excitation	External airborne excitation at modest level	No bolting or hard-coupling to house joists in Rev B. Keep loose objects clear and monitor floor/wall vibration.
Sound exposure	Start quiet; increase only as needed	Use hearing protection and measured SPL; stop for pain, pressure, dizziness, or rattling building components.

7 Commissioning and measurement sequence

Test	Method	Pass / abort criterion
T0 — Inspection	Photograph joints; torque-mark nuts; confirm vessel clearance and guards.	No cracks, rocking, loose hardware, sharp edges, or unguarded string ends.
T1 Operating-load observation	After the load review, observe at the reviewed operating tensions for 30 min; record loaded deflection and unloaded recovery.	No progressive movement or slip; satisfy the reviewed deflection limits. This observation is not a proof-load certification.
T2 — Tip/restraint	First calculate the unrestrained stability margin. Then use a documented controlled restraint test that protects the vessel and operator.	Meet the reviewed tip/restraint criteria. The previous automatic 10° tilt instruction is replaced by assessment before testing.
T3 — Tap map	Damp strings. Tap base, vessel, spine, and top separately with repeatable excitation; retain fixed sensor/reference positions.	Repeatable peaks with raw recordings; distinguish room response and mounting artifacts from component modes.
T4 — Individual strings	Pluck each string at a marked position and initial displacement. Record fundamental, harmonics, and a defined 10 dB decay interval.	No strike or slip. Report pitch drift and repeat variation; fit decay only above the noise floor and without unresolved beating.
T5 Transfer response	Measure excitation or retain a simultaneous reference. Use fixed or calibrated sensor mounts at spine, vessel, top, and base.	Report frequency-dependent transfer functions only with measured input; otherwise label setup-specific response ratios.
T6 — Airflow map	Use yarn tufts around top gap with no powered exciter; video from fixed angle. Then repeat with one guarded external fan.	Tufts remain clear of strings/hardware; no debris ingestion or unstable vibration.

Test	Method	Pass / abort criterion
T7 — Room sweep	Place an external speaker/subwoofer ≥ 3 ft away; sweep 20–200 Hz slowly at low SPL, strings initially damped.	Use the predeclared SPL ceiling; stop for structural rattles, visible instability, or discomfort. Room resonances are not field-model evidence.
T8 — Optional Phase 2	Only after T0–T7 pass: attach ≤ 30 W low-voltage exciter to base, fuse it, repeat T3–T7 at minimum drive.	No temperature rise $> 10^\circ\text{C}$ at exciter mount after 10 min; no loosening or new contact.

Decay and transfer interpretation

For a measured excitation force F and sensor output Y , report $H(f) = Y(f)/F(f)$ with units and mounting conditions. Without measured force, report a response ratio to the retained reference. Disable automatic gain changes when possible and document any processing.

For one isolated lightly damped mode with amplitude proportional to $\exp(-\pi f_0 t/Q)$, a 10 dB amplitude drop gives:

$$Q = \frac{2\pi f_0 t_{10}}{\ln 10} \approx 2.72875 f_0 t_{10}$$

State the fitted frequency band, start/end times, background noise, and uncertainty. Beating, clipping, changing gain, and multi-mode decay invalidate a single-mode Q estimate. Ordinary acoustic resonance is not evidence that this apparatus realizes the TOROIDAL gauge model or its prescribed flow field.

8 Evidence log

For each run record the configuration ID and revision, date/time, string identities and individual pitches/tensions, sensor identity and mounting, excitation method and reference level, sample rate and gain settings, room configuration, peak frequencies, decay-fit interval, temperature, anomalies, and stop reason. Save raw recordings. Repeat each measurement at least three times as an initial repeatability check; do not treat those repeats as a universal statistical pass criterion. Change one experimental factor per comparison.

9 Mechanical comparison and cultural analogy

Interpretation boundary. The analogy column is cultural/design language only. It supplies no evidence of acoustic, electrical, structural, physiological, or environmental mechanism.

Physical feature	Mythic analogy only	Engineering interpretation / boundary
Long central spine + resonating enclosure	Veena / vina	Useful visual family: long neck and resonant body; actual mechanism is ordinary string and enclosure vibration.
Five vertical strings	Lyre / harp	Useful layout family: individually tuned strings; actual behavior depends on length, tension, mass per length, anchors, and damping.
Ordered spacing and removable top	Apollo's lyre	Proportion/order analogy only; no solar, healing, or field claim.

Physical feature	Mythic analogy only	Engineering interpretation / boundary
Environmental listening / room sweep	Dagda's harp; Orpheus's lyre	Narrative analogy for music affecting a setting; measured room modes and structural coupling are the only testable subjects here.
Open top gap and airflow mapping	Gandharvas / Aeolian-harp family	Wind/breath analogy only; airflow is visualized with tufts and measured as ordinary fluid motion.
Optional unenergized copper ornament	Hermes/caduceus imagery	Decorative staff/serpent analogy only. No coil, antenna, charge-focusing, or electromagnetic-mechanism claim.

10 Exclusions and revision control

- No Tesla coil, primary/secondary HV winding, capacitive terminal, corona, exposed energized metal, or predicted electromagnetic “field pattern.”
- No claim that a cymbal “focuses charge,” that a helix improves field stability, or that mythology identifies a mechanism.
- No structural attachment to joists, walls, or building framing except a slack anti-tip safety restraint to a wall stud.
- No smoke, flame, incense, salt, conductive liquids, dust, or aerosol near the assembly.
- Record every change in the revision log. Changes to string family, load path, powered excitation, structural coupling, or conductive elements require reassessment of the affected limits and measurements. Rev B contains multiple documented corrections; within a comparison experiment, vary one factor at a time.

11 Build decision and completion record

Retain the 42 in height, 16 in base, and 32 in speaking length only as provisional planning dimensions. Release a dimensioned cutting layout after the vessel profile, string schedule, anchor connections, frame loading, support polygon, and restraint have been reviewed together. A 12 in clear vessel diameter by itself is not a fit certificate. No construction, load, or acoustic result is certified by this revision.

12 Revision record and calculation sources

Rev B is a design-document correction to Rev A dated 13 August 2026. It removes the build-ready label, fixes the anchor/clearance conditions, separates planning limits from verified capacity, replaces the automatic tip test with assessment first, and makes the acoustic comparison reproducible. The original passive baseline, independent vessel support, cultural-analogy boundary, and powered-excitation exclusions are retained. Comments mark the main local corrections.

Source file SHA-256: bdd4219835f3d3655eed8f816067623e4b548d8a3ca45ea40b72e1141100ffed

UNSW School of Physics, Strings, standing waves and harmonics. Ideal string-frequency relation. [Read source](#)

Baker and Haynes, Engineering Statics, Statically Equivalent Systems. Eccentric forces and equivalent moments. [Read source](#)

Baker and Haynes, Engineering Statics, Center of Gravity. Resultant weight and moment balance. [Read source](#)

Anchor fit, clearance, support-polygon examples, and the decay conversion are calculations stated in this revision. They do not substitute for the missing material, connection, and measurement inputs. No load test, acoustic run, or final fabrication release has been completed by preparing this document.